

Cloud Deployment Options

HRMS - SiteVigil Disciplinary Monitoring System
Multi-Branch Architecture Guide

1. VPS (Virtual Private Server) - DigitalOcean / Linode / Vultr

How it works: You rent a virtual machine (Linux/Ubuntu) from a cloud provider. You install Node.js, PostgreSQL, and PM2 process manager. Upload your application code, point a custom domain, and secure it with free SSL via Let's Encrypt. Branch employees access the system via <https://yourdomain.com>.

PROS

- Full control - complete root access to the server
 - Fixed pricing - predictable monthly cost, no surprise bills
 - Scalable - upgrade RAM/CPU/storage anytime
 - Can host multiple apps on one server
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CONS

- Requires Linux/server administration skills (SSH, firewall, nginx)
- Manual setup - you configure everything from scratch
- You handle backups, security patches, and uptime monitoring
- No graphical interface - all command-line based

Estimated Monthly Cost: \$5 - \$12 / month

2. PaaS (Platform as a Service) - Render / Railway

How it works: Connect your GitHub repository to Render or Railway. They automatically detect your Node.js app, build it, and deploy it. They provide a public URL (e.g., yourapp.onrender.com), handle SSL certificates automatically, and offer built-in PostgreSQL. No server management needed.

PROS

- Zero server management - just git push to deploy
 - Auto HTTPS/SSL - secure connection out of the box
 - Free tier available - enough for demo and small teams
 - Built-in PostgreSQL, logging, and monitoring dashboards
 - Automatic scaling when traffic increases
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CONS

- Free tier sleeps after inactivity (slow restart on first visit)
- Vendor lock-in - harder to migrate to another platform
- Free tier has CPU/month and bandwidth limits
- Becomes expensive at scale (\$20+/mo for decent resources)

Estimated Monthly Cost: \$0 (free tier) - \$7 (starter)

3. Supabase (Backend-as-a-Service)

How it works: Supabase is an open-source Firebase alternative. It provides hosted PostgreSQL, built-in authentication, file storage, and auto-generated REST API. You rewrite your app to call Supabase APIs instead of SQLite. Frontend can be hosted for free on Vercel or Netlify.

PROS

- Very generous free tier (500MB database, 2GB storage, 50k rows)
 - Built-in authentication - email, Google, Facebook, etc.
 - Real-time subscriptions - data updates push to UI automatically
 - Auto backups, graphical dashboard, SQL editor in browser
 - Open-source - you can self-host if needed later
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CONS

- Requires code rewrite - replace SQLite with Supabase API calls
- Free tier row limit (50,000 rows across all tables)
- API rate limits and bandwidth caps on free plan
- Your data resides on their cloud infrastructure

Estimated Monthly Cost: \$0 (free tier) - \$25 (pro)

RECOMMENDATION: PaaS (Render or Railway)

For your current stage (system proposal period), Render or Railway is the best choice because:

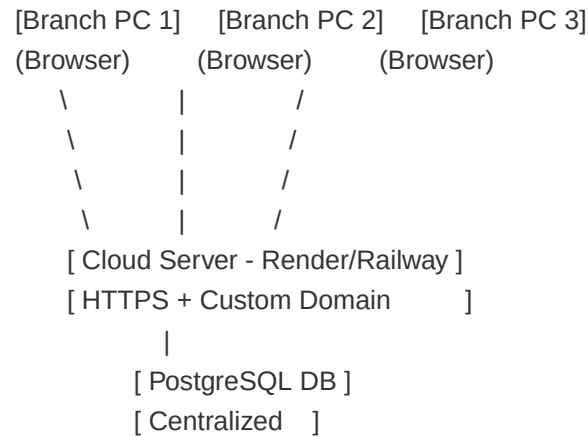
- No server knowledge needed - just git push and it deploys
- Free tier works immediately for client demos and presentations
- Professional HTTPS URL instills client confidence
- Easy to upgrade later as the system grows
- Minimal code changes needed (SQLite to PostgreSQL migration only)

Migration Path: SQLite to PostgreSQL

The current system uses better-sqlite3 (local file-based database). To deploy on cloud, we migrate to PostgreSQL:

- Replace better-sqlite3 with pg (node-postgres) driver
- Convert SQLite-specific queries to PostgreSQL-compatible syntax
- Set up connection pooling for performance
- Use environment variables for database credentials
- Deploy and test with a staging environment first

Proposed Cloud Architecture



Branch Isolation Strategy

- Each branch has a branch_id field in all major tables
- Middleware automatically filters data by branch on every query
- Branch admins only see their own branch data
- Main office (admin role) sees all branches
- No VPN or networking needed - just internet access

Next Steps When Ready

- I will migrate the database from SQLite to PostgreSQL
- Set up environment variables for security (database credentials, JWT secret, SMTP)
- Deploy to Render/Railway free tier in under 30 minutes
- You get a live HTTPS URL to share with your client
- Configure automatic daily backups and uptime monitoring